School Assistant (Mathematics) - Mock Test 102

Total Marks: 80 | Total Questions: 80 | Duration: 150 Mins | Medium: English

--- General Knowledge & Current Affairs ---

Q1. Under which Article of the Constitution of India is the Right to Free and Compulsory Education (RTE) guaranteed as a Fundamental Right for children aged 6 to 14?
(A) Article 19 (B) Article 21A (C) Article 32 (D) Article 45

Q2. The 73rd Constitutional Amendment Act, 1992, which granted constitutional status to Panchayati Raj Institutions, added which Schedule to the Constitution of India?
(A) 10th Schedule (B) 11th Schedule (C) 12th Schedule (D) 9th Schedule

Q3. Which of the following is the longest river flowing through the Deccan Plateau in peninsular India?
(A) Krishna (B) Narmada (C) Godavari (D) Cauvery

Q4. The sparkling of a diamond and the transmission of light through optical fiber cables are primarily based on which optical phenomenon?
(A) Total Internal Reflection (B) Diffraction of light (C) Polarization (D) Atmospheric scattering

Q5. India's Chandrayaan-3 mission landed on the lunar South Pole on August 23, 2023. What is the official name designated for the touchdown landing point?
(A) Tiranga Point (B) Shiv Shakti Point (C) Vikram Sthal (D) Atal Point

Q6. The Ramsar Convention, signed in 1971 in Ramsar, Iran, is an international treaty specifically dedicated to the conservation and sustainable use of:
(A) Tropical Rainforests (B) Wetlands (C) Ozone Layer (D) Endangered Desert Mammals

Q7. The PM POSHAN Scheme (formerly known as the National Programme of Nutritional Support to Primary Education or Mid-Day Meal Scheme) provides hot cooked meals to children studying in:
(A) Pre-primary to Class VIII in government and government-aided schools (B) Class IX to XII only (C) Private unaided higher secondary schools (D) University residential colleges

Q8. In which historic session of the Indian National Congress held in December 1929 was the resolution for 'Purna Swaraj' (Complete Independence) formally passed under the presidency of Jawaharlal Nehru?
(A) Karachi Session (B) Lahore Session (C) Lucknow Session (D) Calcutta Session

Q9. The 13th-century Ramappa Temple (Kakatiya Rudreshwara Temple), inscribed as a UNESCO World Heritage Site, is renowned for its sandbox technology and lightweight floating bricks. In which district is it situated?
(A) Mulugu District (Telangana) (B) Warangal Urban (C) Khammam (D) Nalgonda

Q10. Under the National Education Policy (NEP) 2020, what is the new pedagogical and curricular restructuring replacing the traditional 10+2 school education system?
(A) 5 + 3 + 3 + 4 (B) 4 + 4 + 4 + 2 (C) 3 + 5 + 4 + 2 (D) 5 + 4 + 3 + 2

--- Perspectives in Education ---

Q11. Which historic British education policy document is widely revered as the 'Magna Carta of English Education in India'?
(A) Macaulay's Minute of 1835 (B) Wood's Despatch of 1854 (C) Hunter Commission of 1882 (D) Sargent Plan of 1944

Q12. The Hartog Committee Report (1929) is best known in the history of Indian education for drawing critical attention to which two systemic defects in primary schooling?
(A) 'Wastage' and 'Stagnation' (B) 'Privatization' and 'Commercialization' (C) 'Vocationalization' and 'Industrialization' (D) 'Centralization' and 'Examination Stress'

Q13. Mahatma Gandhi propounded the scheme of Basic Education (Nai Talim) at the All India National Education Conference held at Wardha in 1937. Under whose chairmanship was the detailed syllabus committee appointed?
(A) Dr. Zakir Hussain (B) Rabindranath Tagore (C) Gopal Krishna Gokhale (D) Maulana Abul Kalam Azad

Q14. "The destiny of India is now being shaped in her classrooms" is the famous opening statement of which landmark education commission report?
(A) Mudaliar Commission (1952-53) (B) Kothari Commission (1964-66) (C) Radhakrishnan Commission (1948-49) (D) National Knowledge Commission (2005)

Q15. 'Operation Blackboard' was a major centrally sponsored scheme launched in 1987 as a direct outcome of which national education initiative?
(A) National Policy on Education (NPE) 1986 (B) Sargent Report 1944 (C) Right to Education Act 2009 (D) National Curriculum Framework 2005

Q16. Under Section 12(1)(c) of the Right of Children to Free and Compulsory Education (RTE) Act, 2009, private unaided schools are mandated to reserve what percentage of Class 1 admissions for children from disadvantaged and economically weaker sections?
(A) 10% (B) 15% (C) 25% (D) 33%

Q17. According to Section 21 of the RTE Act, 2009, what proportion of the members of a School Management Committee (SMC) must consist of parents or guardians of students studying in the school?
(A) At least 50% (B) At least 60% (C) At least 75% (three-fourths) (D) 100%

Q18. Which of the following is NOT one of the five core guiding principles formulated in the National Curriculum Framework (NCF) 2005?
(A) Connecting knowledge to life outside the school (B) Ensuring that learning shifts away from rote methods (C) Making examinations more flexible and integrated into classroom life (D) Encouraging strict memorization of textbook answers for board examinations

Q19. Which statutory apex body established under an Act of Parliament in 1993 oversees standards, regulations, and qualifications for teacher education institutions throughout India?
(A) National Council for Teacher Education (NCTE) (B) National Council of Educational Research and Training (NCERT) (C) University Grants Commission (UGC) (D) Central Board of Secondary Education (CBSE)

Q20. As per the Norms and Standards for a School specified in the Schedule of the RTE Act, 2009, what is the mandated Pupil-Teacher Ratio (PTR) for primary schools (Classes I to V) up to 60 admitted children?
(A) 25 : 1 (B) 30 : 1 (Two teachers for up to 60 children) (C) 35 : 1 (D) 40 : 1

--- Mathematics Content ---

Q21. If a positive integer a is divided by 5, the possible remainders according to Euclid's Division Lemma are:
(A) 0 to 4 (B) 1 to 5 (C) 0 to 5 (D) -1 to 4

Q22. For two numbers with HCF = 14 and LCM = 210. If one number is 42, find the other number.
(A) 70 (B) 56 (C) 84 (D) 112

Q23. The rationalising factor of (sqrt(5) + sqrt(2)) is:
(A) sqrt(5) - sqrt(2) (B) sqrt(5) + sqrt(2) (C) 5 - 2 (D) sqrt(10)

Q24. If log_2(x) = 3, then the value of x is:
(A) 8 (B) 6 (C) 5 (D) 2

Q25. The modulus of the complex number z = 3 + 4i is:
(A) 5.0 (B) 7 (C) 25 (D) 4

Q26. If set A has 5 elements, what is the total number of proper subsets of A?
(A) 31 (B) 32 (C) 30 (D) 10

Q27. For two finite sets A and B, n(A) = 22, n(B) = 17, and n(A union B) = 36. What is n(A intersect B)?
(A) 3 (B) 39 (C) 36 (D) 14

Q28. A function f: R -> R defined by f(x) = 2x + 5 is:
(A) Bijective (One-one and Onto) (B) One-one but not Onto (C) Onto but not One-one (D) Neither One-one nor Onto

Q29. If the price of sugar increases by 20%, by what percentage must consumption be reduced so that the expenditure remains unchanged?
(A) 16.67% (B) 20% (C) 25.0% (D) 15%

Q30. A shopkeeper marks an article at Rs. 1200 and offers a discount of 10%. If he makes a profit of 20%, what was the cost price (CP)?
(A) Rs. 900 (B) Rs. 960 (C) Rs. 900 (D) Rs. 1020

Q31. The difference between Compound Interest and Simple Interest on a principal of Rs. 10000 for 2 years at 7% per annum is:
(A) Rs. 49 (B) Rs. 700 (C) Rs. 98 (D) Rs. 70

Q32. A can complete a piece of work in 10 days and B can do it in 15 days. In how many days can they complete it together?
(A) 6.0 days (B) 25 days (C) 12 days (D) 8 days

Q33. When P(x) = x^3 - 2x^2 + 4x - 5 is divided by (x - 2), what is the remainder?
(A) 3 (B) 8 (C) 0 (D) -5

Q34. For what value of k is (x - 1) a factor of polynomial P(x) = 2x^3 - k x^2 + 5x - 3?
(A) 4 (B) 0 (C) 0 (D) 1

Q35. The system of linear equations 2x + 3y = 7 and 4x + 6y = 15 has:
(A) No solution (Inconsistent, parallel lines) (B) A unique solution (C) Infinitely many solutions (D) Exactly two solutions

Q36. The roots of the quadratic equation x^2 - 5x + 6 = 0 are:

(A) Real and distinct (B) Real and equal (C) Complex conjugates (D) Pure imaginary

Q37. If alpha and beta are the roots of 3x^2 - 9x + 5 = 0, then the value of (alpha + beta) is:
(A) 3.0 (B) 5/3 (C) -3.0 (D) -5/3

Q38. What is the total number of terms in the binomial expansion of (x + a)^12?
(A) 13 (B) 12 (C) 11 (D) 24

Q39. The sum of all binomial coefficients C(n, 0) + C(n, 1) + ... + C(n, n) for n = 7 is equal to:
(A) 128 (B) 49 (C) 14 (D) 64

Q40. In an Arithmetic Progression with first term a = 4 and common difference d = 3, what is the 20th term?
(A) 61 (B) 64 (C) 60 (D) 57

Q41. The sum of an infinite geometric series with first term a = 6 and common ratio r = 1/2 is:
(A) 12 (B) 6 (C) 3.0 (D) 24

Q42. For two distinct positive real numbers a and b, which inequality strictly holds between their Arithmetic Mean (AM), Geometric Mean (GM), and Harmonic Mean (HM)?
(A) AM > GM > HM (B) HM > GM > AM (C) GM > AM > HM (D) AM = GM = HM

Q43. In triangle ABC, DE is parallel to BC with D on AB and E on AC. If AD = 2 cm, DB = 4 cm, and AE = 3 cm, find the length of EC.
(A) 6.0 cm (B) 6 cm (C) 4 cm (D) 8 cm

Q44. The corresponding sides of two similar triangles are in the ratio 3 : 4. The ratio of their areas is:
(A) 9 : 16 (B) 3 : 4 (C) 27 : 64 (D) 6 : 8

Q45. In a cyclic quadrilateral ABCD, if angle A = 80 degrees, then the measure of the opposite angle C is:
(A) 100 degrees (B) 80 degrees (C) 90 degrees (D) 135 degrees

Q46. A tangent PT is drawn from an external point P to a circle with centre O and radius 5 cm. If the distance OP is 13 cm, find the length of the tangent PT.
(A) 12 cm (B) 10 cm (C) 14 cm (D) 8 cm

Q47. According to the Alternate Segment Theorem, the angle between a tangent to a circle and a chord drawn through the point of contact is equal to:
(A) The angle subtended by the chord in the alternate segment (B) Twice the angle subtended by the chord at the centre (C) Complement of the angle at the centre (D) 90 degrees

Q48. An equilateral triangle has side a = 6 cm. What is its exact area?
(A) 15.59 cm^2 (B) 36 cm^2 (C) 5.2 cm^2 (D) 18 cm^2

Q49. Find the area of a sector of a circle with radius r = 7 cm and central angle theta = 60 degrees. (Use pi = 22/7)
(A) 25.67 cm^2 (B) 154.0 cm^2 (C) 22 cm^2 (D) 44 cm^2

Q50. A right circular cylinder has base radius r = 7 cm and height h = 12 cm. Its volume is: (Use pi = 22/7)
(A) 1848 cm^3 (B) 528 cm^3 (C) 264 cm^3 (D) 616 cm^3

Q51. The surface area of a sphere of radius r = 7 cm is: (Use pi = 22/7)
(A) 616 cm^2 (B) 308 cm^2 (C) 154 cm^2 (D) 1232 cm^2

Q52. If sec(theta) + tan(theta) = 2, then the value of sec(theta) - tan(theta) is:
(A) 1 / 2 (B) -(2) (C) 2 (D) 1

Q53. A vertical pole of height 17.32 m casts a shadow of length 10 m on the ground. The angle of elevation of the sun is:
(A) 60 degrees (B) 30 degrees (C) 45 degrees (D) 75 degrees

Q54. If sin(A) = 3/5 (where A is acute), what is the value of sin(2A)?
(A) 24/25 (B) 12/25 (C) 7/25 (D) 16/25

Q55. The value of sin(arcsin(x) + arccos(x)) for any x in [-1, 1] is:
(A) 1 (B) 0 (C) 1/2 (D) x

Q56. The distance between points P(1, 2) and Q(4, 6) is:
(A) 5 units (B) 7 units (C) sqrt(7) units (D) 25 units

Q57. The slope of the straight line 3x + 4y + 9 = 0 is:
(A) -0.75 (B) 0.75 (C) -4/3 (D) -9/4

Q58. The perpendicular distance from origin (0, 0) to the line 3x + 4y - 10 = 0 is:
(A) 2.0 units (B) 10 units (C) 5 units (D) 2.5 units

Q59. The eccentricity (e) of a parabola is:
(A) e = 1 (B) e < 1 (C) e > 1 (D) e = 0

Q60. If matrix A has order 2 x 3 and matrix B has order 3 x 4, then the order of the product AB is:

(A) 2 x 4 (B) 3 x 3 (C) 4 x 2 (D) Undefined

Q61. If A is a 3 x 3 square matrix with det(A) = 2, what is det(2A)?
(A) 16 (B) 4 (C) 8 (D) 12

Q62. For a moderately skewed distribution, if Mean = 22 and Median = 24, the empirical Mode is:
(A) 28 (B) 26 (C) 46 (D) 23

Q63. If A and B are independent events with P(A) = 0.3 and P(B) = 0.4, then P(A intersect B) is:
(A) 0.12 (B) 0.7 (C) 0.1 (D) 0.7

Q64. In standard ANSI flowcharting conventions, which geometric shape represents a conditional decision-making branch with multiple outcomes?
(A) Diamond (Rhombus) (B) Rectangle (C) Parallelogram (D) Oval

--- Teaching Methodology ---

Q65. Which characteristic most distinctively defines the nature and epistemic structure of mathematics?
(A) Abstractness and Deductive logical consistency (B) Empirical observation and experimental laboratory trials (C) Subjective interpretation and emotional appreciation (D) Unsystematic trial and accidental discovery

Q66. Why is 1729 celebrated in mathematics as the 'Ramanujan-Hardy Number'?
(A) It is the smallest number expressible as the sum of two cubes in two different ways (1^3 + 12^3 and 9^3 + 10^3) (B) It is the largest prime factor in partition theory (C) It represents the exact ratio of the golden rectangle (D) It is the first amicable number discovered by Fermat

Q67. In the revised Bloom's Taxonomy of Educational Objectives (Anderson & Krathwohl, 2001), what is the highest cognitive process level?
(A) Creating (B) Evaluating (C) Analyzing (D) Applying

Q68. When a student writes a mathematical proof showing that the sum of angles of a triangle is 180 degrees, this behavioral outcome corresponds primarily to which objective?
(A) Analysis and Logical Reasoning (B) Simple Recall (C) Affective Characterization (D) Imitation

Q69. A mathematics teacher presents five different triangles, asks students to measure their angles, find their sum, and conclude that the sum of angles in any triangle is 180 degrees. Which teaching method is being used?
(A) Inductive Method (B) Deductive Method (C) Lecture Method (D) Synthetic Method

Q70. The pedagogical approach that proceeds systematically from 'Unknown to Known' by decomposing a complex problem into simpler components is the:
(A) Analytic Method (B) Synthetic Method (C) Drill Method (D) Rote Memorization Method

Q71. Who is the originator of the 'Heuristic Method' of teaching, derived from the Greek word 'heurisko' meaning 'I find'?
(A) Professor H.E. Armstrong (B) John Dewey (C) William Kilpatrick (D) B.F. Skinner

Q72. In his classic work 'How to Solve It', mathematician George Polya formulated four distinct phases of mathematical problem solving. What is the fourth and final step?
(A) Looking Back (Reflection and Verification) (B) Understanding the Problem (C) Devising a Plan (D) Carrying out the Plan

Q73. "A project is a wholehearted purposeful activity proceeding in a social environment" is the definition propounded by:
(A) William Heard Kilpatrick (B) John Dewey (C) J.A. Stevenson (D) Ballard

Q74. Which of the following represents the correct sequential order of the classical Herbartian formal steps in lesson planning?
(A) Preparation -> Presentation -> Association -> Generalisation -> Application -> Recapitulation (B) Application -> Generalisation -> Presentation -> Preparation -> Association (C) Generalisation -> Association -> Preparation -> Presentation -> Application (D) Presentation -> Application -> Preparation -> Generalisation -> Association

Q75. According to Edgar Dale's Cone of Experience, which learning experience lies at the base of the cone and offers the most concrete learning foundation?
(A) Direct Purposeful Experiences (B) Verbal Symbols (C) Visual Symbols (Diagrams) (D) Radio and Recordings

Q76. What is the primary instructional purpose of establishing a Mathematics Laboratory in secondary schools?
(A) To provide experiential, manipulative-based learning and verify abstract mathematical concepts concretely (B) To replace all classroom textbooks entirely (C) To administer high-stakes annual board examinations (D) To force students into silent independent memorization

Q77. A student experiences persistent difficulty with number sense, basic arithmetic calculations, and understanding mathematical symbols despite normal intelligence. This learning difficulty is termed:
(A) Dyscalculia (B) Dyslexia (C) Dysgraphia (D) Apraxia

Q78. What is the indispensable logical sequence between Diagnostic Testing and Remedial Teaching in mathematics education?
(A) Diagnostic testing identifies specific learning gaps first; Remedial teaching provides corrective instructional measures (B)

Remedial teaching is conducted first; Diagnostic testing is given only to pass students (C) Both are given simultaneously as a punitive grading exercise (D) Remedial teaching replaces formative assessment completely

Q79. In constructing a balanced Achievement Test in Mathematics, what is the fundamental purpose of preparing a 'Blueprint'?
(A) To ensure balanced weightage to instructional objectives, content areas, and question formats (B) To print answer keys in advance for grading convenience (C) To ensure that all questions are of the hard difficulty level (D) To test only rote recall of mathematical formulas

Q80. If an achievement test in mathematics consistently yields identical scores when administered repeatedly to the same group of students under unchanged conditions, the test is said to possess high:
(A) Reliability (B) Validity (C) Subjectivity (D) Usability